Application Flow & Architectural Workflow Document

System Scope: Dual-Surface Higher-Education Dormitory Operations, Resident Lifecycle & Facility Triage



Figure 1.0: RÉSIVO Dual-Surface Operations Ecosystem (Student Web Portal ↔ Central Supabase Hub ↔ Staff Operations Console)

1. High-Level Operations Overview

🌐 Resident Web Portal

Platform: Astro 5 + Tailwind CSS v4
Mobile-first portal for students: self-service registration, card ID verification, digital room passes, and maintenance ticketing with camera photo uploads.

🖥️ Staff Operations Console

Platform: PySide6 (Qt 6) + Python 3.11
Administrative desktop workstation: 2-second background sync worker, roster matching, room allocations, desk inquiry references, and async photo viewer.

🤖 AI Operations Assistant

Platform: Cloudflare Workers (MCP)
Direct Model Context Protocol integration with ChatGPT: natural language occupancy queries and ticket status lookups with strict authorization.

2. Architectural Interaction Model

CHANNEL	SOURCE → TARGET	CADENCE / PROTOCOL	SAFEGUARDS & INVARIANTS
Live State Polling	Staff Desktop → PostgreSQL	2000ms Interval (`LiveSyncWorker`)	Differential key check (`old_keys == new_keys`) avoids redraw flicker; preserves active filter tabs & search text.

CHANNEL	SOURCE → TARGET	CADENCE / PROTOCOL	SAFEGUARDS & INVARIANTS
Photo Evidence	Web Portal → Supabase Storage	On-demand Multipart Upload	Restricted to `maintenance-photos` bucket; fetched via multi-threaded background worker on desktop.
Dossier Deletion	Staff Desktop → Postgres RPC	Administrative Confirmation	Pending applications locked against deletion; only active/rejected students can be pruned; cascades room assignments.

3. End-to-End Residency Process Sequence

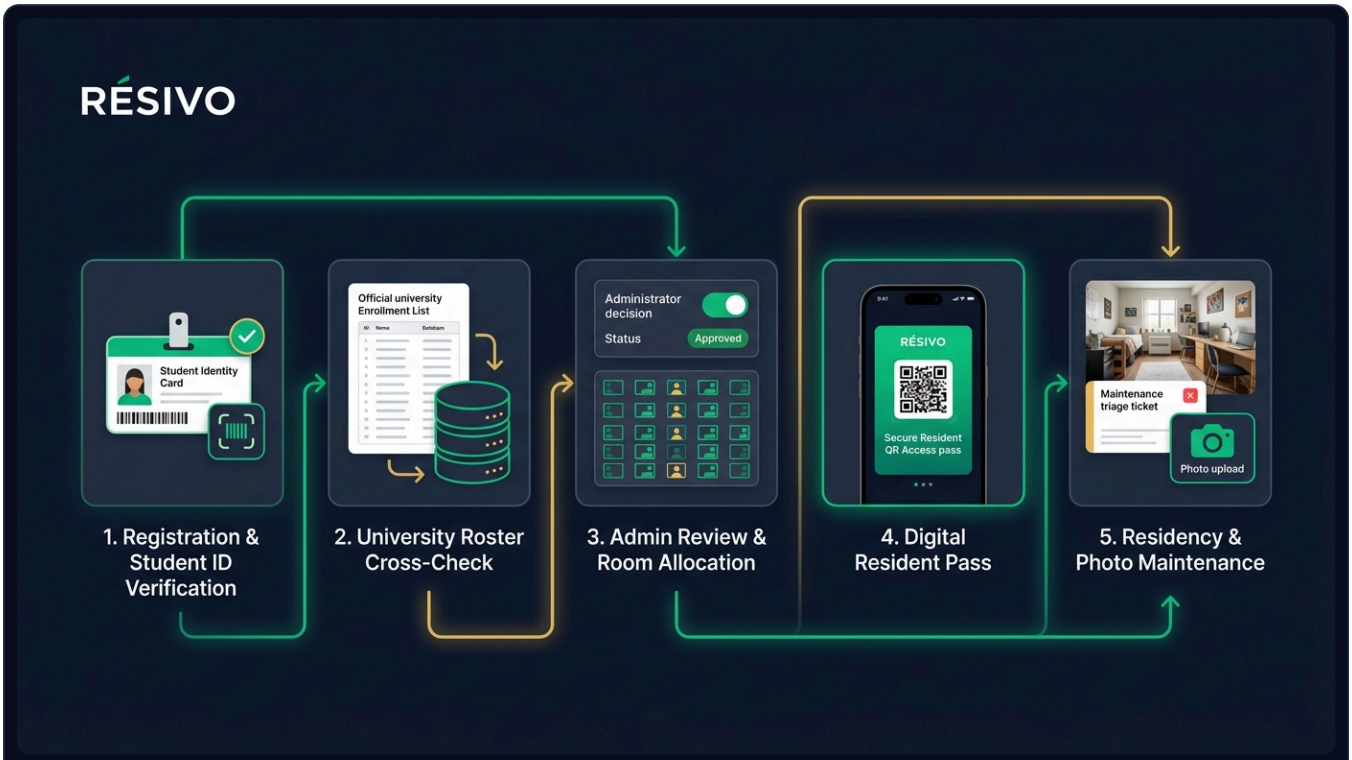


Figure 2.0: The 5-Phase Student Housing Journey from Registration to Triage Resolution

1 **Phase 1: Student Registration & Academic Identity Submission**

Student accesses `/register` on mobile or desktop web. Submits national student card ID (e.g. `202437010001`), full legal name, wilaya, gender, faculty, phone, and password. System authenticates user via Supabase Auth and inserts application with status **PENDING**.

2 **Phase 2: Automated Cross-Referencing against Official University Roster**

The system queries `official_residents` table (imported from university enrollment files). If the card ID exists, the applicant is tagged **✓ MATCHED** with verified faculty details. If absent, it is flagged with **⚠ NOT IN ROSTER**.

3 **Phase 3: Administrative Triage & Room Allocation (`PendingPage`)**

Housing staff filters by `Pending Applications`. Evaluates academic dossier and roster flag. If approved, opens the Room Allocation modal, which enforces block gender segregation (Block A = Male, Block B = Female) and validates capacity (≤ 2). Application becomes **ACTIVE**.

4 **Phase 4: Digital Resident Pass Issuance & Dormitory Check-In**

Upon approval, the student's dashboard unlocks their **Digital Resident Pass** displaying assigned Block, Floor, Room Number, and matriculation barcode. Locked services (Maintenance & Transfer) immediately activate.

5 **Phase 5: Dormitory Living & Photo Maintenance Triage**

Residents report facility faults by selecting a category and snapping a photo. Staff desktop receives real-time notification, inspects full-resolution image with the async photo viewer, and updates status to **IN PROGRESS** and **RESOLVED**.

4. Application Triage & Desk Inquiry Decision Flow

The housing department console enforces institutional verification protocols during application processing:

Decision A: Application Approval

1. Staff selects applicant from `Pending Applications`.
2. Inspects roster match badge: `✓ Matched` or `⚠ Not in Roster`.
3. Clicks **Approve & Assign Room**.
4. System filters rooms by student gender & remaining capacity.
5. Staff confirms target room; database executes atomic update:
   `UPDATE students SET room_id = target_room, status = 'active', reviewed_at = NOW();`
6. Student's digital pass is instantly generated and visible on web.

Decision B: Application Rejection

1. Staff clicks **Reject Application**.
2. System prompts for official administrative justification via `QInputDialog`.
3. **Mandatory Validation:** Empty reasons are strictly blocked with `QMessageBox.warning`.
4. Staff enters formal note (e.g. *"Missing residence slip"*).
5. Calls PostgreSQL RPC `reject_student`.
6. Application status becomes `REJECTED`.

Front Desk Inquiry Support Reference

When a rejected applicant inquires at the housing reception desk, the officer filters by `Rejected Applications`. The detail inspector displays:

```
⚠ REJECTION REASON (Desk Inquiry Reference):
"Incomplete academic file: missing residence registration slip"
Reviewed: 2026-10-12 14:30:15
Student: Mohamed Belkacem (Card ID: 202437010003)
```

5. Multi-Tier Deletion Safety Matrix

TARGET RECORD	APPLICATION STATE	DELETION BUTTON STATE	BACKEND DEFENSE MECHANISM
Student Dossier	`PENDING`	**Disabled (Greyed Out)** Tooltip: *Pending applications cannot be deleted.*	Database RPC `delete_student` checks `status = 'pending'` → raises exception. RLS blocks delete.
Student Dossier	`APPROVED`	**Enabled (Red / Danger)** Prompts `QMessageBox.question`	Clears room assignment; cascades maintenance reports and room requests; deletes auth user.
Student Dossier	`REJECTED`	**Enabled (Red / Danger)** Prompts `QMessageBox.question`	Permanently removes rejected student dossier and associated credentials.
Staff Account	Logged-in Self	**Disabled (Greyed Out)** Tooltip: *You cannot delete your own staff account.*	Edge Function checks `id === caller.id` → HTTP 400. RPC checks `p_staff = auth.uid()`.

6. Maintenance Ticketing & High-Resolution Photo Viewer

A **Resident Fault Submission Flow**
Resident selects fault category (`Plumbing` , `Electricity` , `Heating` , `Carpentry` , `General`), provides problem details, and uploads a photo. Image is sent to Supabase Storage with signed token. Ticket state set to PENDING .

B **Asynchronous Desktop Inspection Flow**
Desktop console displays ticket with thumbnail loaded on a non-blocking background `QThread` (`PhotoFetchWorker`). Facility officer clicks thumbnail to launch `PhotoViewerDialog` featuring zoom controls (25% to 400%), fit-to-screen scaling, and clean thread cleanup on exit.

C **Resolution & Triage Lifecycle**
Officer advances ticket stage: PENDING → IN PROGRESS → RESOLVED , appending internal technician notes. Student portal updates dynamically without page reloads.

7. Room Transfer & Adjudication Flow

1. **Locked State Enforcement:** Only residents with status `active` can access the transfer docket; pending and rejected students are locked out.
2. **Target Room Selection:** Resident selects target dormitory block and available room matching their gender, entering written motive.
3. **Housing Staff Review:** Officer reviews justification in `RoomChangesPage` .
4. **Atomic Transfer:** If approved, the system updates the resident's `room_id` , updates request status to `approved` , and recalculates block occupancy immediately.

8. Model Context Protocol (MCP) AI Query Flow

🤖 AI Operations Assistant Architecture

Housing administrators can query real-time dormitory statistics using natural language through ChatGPT Plus/Pro:

- **Query:** *"Show current occupancy rates across all blocks and list pending plumbing tickets."*
- **Flow:** ChatGPT → Cloudflare Worker (`/mcp`) → Bearer Token Validation → Supabase SQL Execution → JSON-RPC Tool Result → ChatGPT Executive Summary.
- **Safety Guard:** Assistant is read-only by default; data modifications require explicit user confirmation.

9. Verification & Certification

SUBSYSTEM	TEST SUITE	COVERAGE METRIC	STATUS
Desktop Console	Pytest 8.3.3 (19 test cases)	100% of status filters, self-deletion, live sync, photo viewer, and window scaling tested	19 / 19 PASS

SUBSYSTEM	TEST SUITE	COVERAGE METRIC	STATUS
Student Web Portal	Vitest 5.0.3	Astro SSR form submission, authentication routes, and profile updates validated	ALL PASS
PostgreSQL Database	Supabase CLI & MCP Runner	Idempotent migrations, RLS policies, and SECURITY DEFINER RPCs verified	VERIFIED